

Document-oriented data stores

These NoSQL key-value data stores are highly structured, and include complex self-defining objects as their values. Many of the popular document databases, such as MongoDB or CouchDB, use JavaScript Object Notation (JSON). CouchDB (couchdb.apache.org) defines a basic key/value storage mechanism to store documents in JSON (JavaScript Object Notation) format. It allows the creation of persistent views on top of this data, similar to database tables which can be queried. The storage engine for CouchDB does support ACID properties, and is intended to be a distributed fault-tolerant system. It has the capabilities to scale down and hence, can be used for mobile computing, efficiently. A similar project is MongoDB (www.mongodb.org), which does not provide ACID properties, but has powerful querying capabilities.

Graph databases

A graph database is a kind of NoSQL database that uses graph structures with nodes, edges, and properties to represent and store information. Nodes represent entities, and properties are information that relate to nodes. Edges connect nodes to nodes/properties representing the relationship. Most of the important information is really stored in the edges. Compared with relational databases, graph databases are often faster for associative data sets, scale more naturally to large data sets, and are more suitable for changing data/evolving schemas. Conversely, relational databases are typically faster at performing the same operation on large numbers of data elements. Examples of graph databases are Neo4j (<http://www.neo4j.org/>) and hypergraphDB (www.hypergraphdb.org/).

Various NoSQL implementations differ considerably on how consistency is achieved, what data types are supported, the structure and nature of how the data is stored, and the support of languages/libraries/interfaces for access. Implementations embrace what is best for them by considering how their targeted class of Web services would need to scale. They also vary in terms of whether the key-values are stored in-memory, on-disk, or both; hierarchically or tabular; duple or triplet; single or multiple value per key, etc. A detailed discussion of NoSQL databases can be found in the paper 'No Relation: The Mixed Blessings of Non-Relational Databases', available at <http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.169.1113>.

What's around the corner for databases?

The question that has frequently been asked is whether the NoSQL movement signifies the end of traditional RDBMS. While the NoSQL movement has helped to answer the initial needs of massive data sets of Web 2.0, traditional OLTP and OLAP enterprise applications still depend on RDBMS. Therefore, we expect that both NoSQL and traditional RDBMS would continue to co-exist for the next decade. Regarding the question on whether today's databases satisfy

the challenges of data management posed by the Cloud and Big Data, we should look back at database history for our answer. Recall that the RDBMS revolution enabled the conversion of business data into information, driving business processes and business intelligence. It is not sufficient for NoSQL databases to support high availability, scalability and low latency needs. They also need to provide state-of-the-art analytics, which can power business intelligence while being elastic to fit the cloud environment. The opinion widely shared by many in the database industry is that, "The real database revolution driven by Big Data and the Cloud is just around the corner." So we do have a lot of interesting innovations to look forward to in the database space over the next few years. 

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